SCHOOL OF ADVANCED SCIENCES

Fall Semester 2024-2025 **Continuous Assessment Test – I**

Programme Name & Branch: B.Tech & CSE **Slot: C1+TC1+TCC1**

Course Name & code: Discrete Mathematics and Graph Theory & BMAT205L

Class Number (s): VL2024250102537, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, ,2547, 2548, 2549, 2552, 2553, 2554, 2555, 2558

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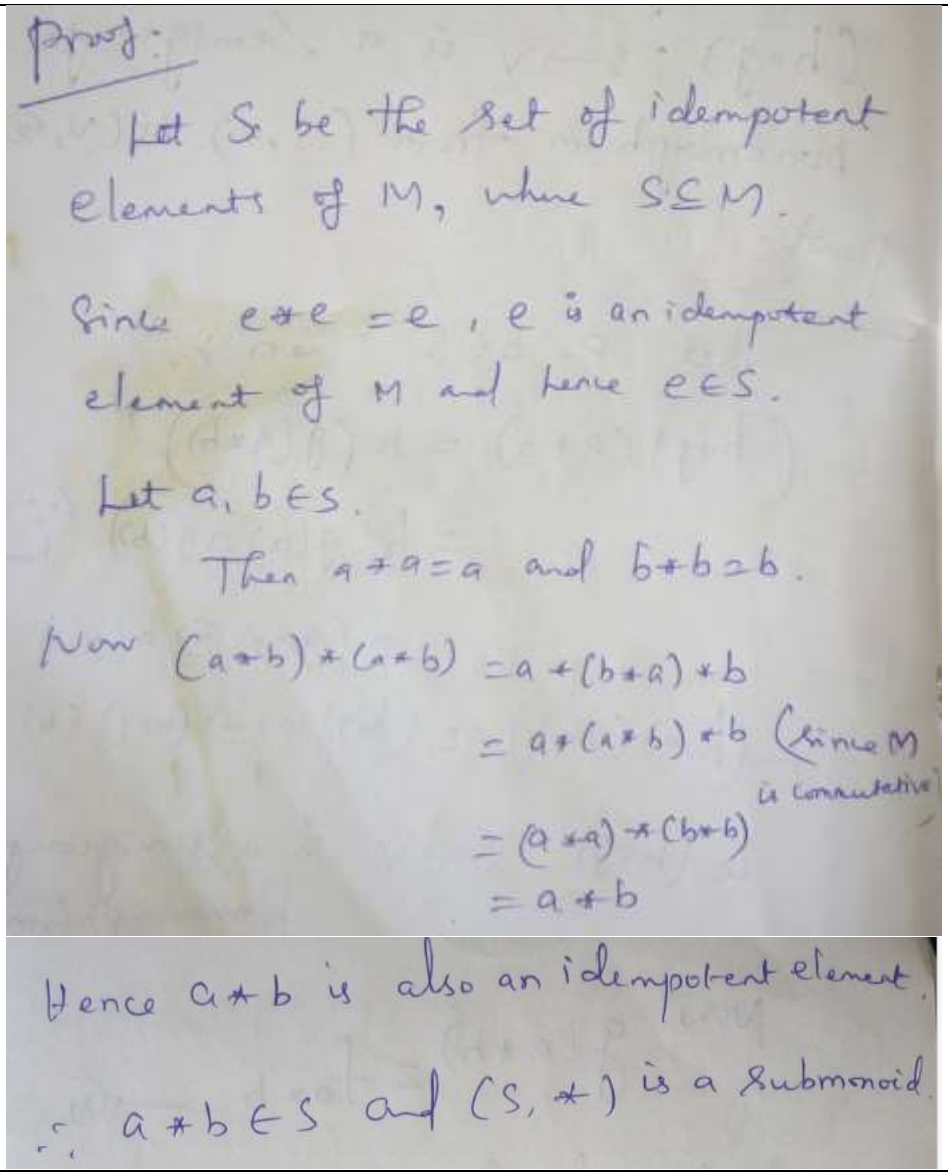
Exam Duration: 90 Min. **Maximum Marks: 50**

General instruction(s): Answer ***ALL*** the following questions.

Q.No.	Question	Max Marks
1.	Find the PDNF and PCNF of $(p \wedge q) \vee (p \wedge r) \vee (q \wedge r)$. (i) By using Truth table (ii) without using truth table.	10
ANS	PDNF: $(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r)$ PCNF: $(\neg p \vee q \vee r) \wedge (p \vee \neg q \vee r) \wedge (p \vee q \vee \neg r) \wedge (p \vee q \vee r)$	
2.	**(i)** Prove the following "All flowers are plants. Sunflower is a flower". "Therefore, Sunflower is a plant". (5M)	10
ANS	P(x) is " x is a plant" a is a constant symbol (Sunflower) F(x) is "x is a flower" The argument is $(\forall x)[F(x) \rightarrow P(x)] \wedge F(a) \rightarrow P(a)$ The proof sequence is as follows: 1. $(\forall x)[F(x) \rightarrow P(x)]$ hyp 2. $F(a)$ hyp 3. $F(a) \rightarrow P(a)$ 1, ui 4. $P(a)$ 2, 3, mp	
	(ii) Show that the following statement is valid. All men are mortal Socrates is a man Therefore, Socrates is a mortal. (5M)	
ANS	H(x): x is a man. M(x): x is a mortal S: Socrates $\forall(x)(H(x) \rightarrow M(x)) \wedge H(S) \Rightarrow M(S)$	

3.	"If I eat spicy food, then I have strange dreams". "I have strange dreams, if there is thunder while I sleep". "I did not have strange dreams". What relevant conclusion can be drawn from the above premises? Construct an argument to obtain your conclusion.	10
ANS	I did not eat spicy food or there was no thunder. • The relevant conclusions are: "I did not eat spicy food" and "There is no thunder while I sleep". • Let the primitive statements be: ▪ *s, 'I eat spicy foods'* ▪ *d, 'I have strange dreams'* ▪ *t, 'There is thunder while I sleep'* • Then the premises are translated as: $s \rightarrow d$, $t \rightarrow d$, and $\neg d$. • And the conclusions: $\neg s$, $\neg t$. • Steps — Reason 1. $s \rightarrow d$ — *premise* 2. $\neg d$ — *premise* 3. $\neg s$ — *Modus Tollens to Steps 1 and 2* 4. $t \rightarrow d$ — *premise* 5. $\neg t$ — *Modus Tollens to Steps 4 and 2.*	
4.	Prove that for any commutative monoid (M, *), the set of idempotent elements of M forms a submonoid.	10
ANS		

<table>
<tr><td>5.</td><td>Check whether the set {5, 15, 25, 35} is a group or not with respect to multiplication module 40.
(i) Form the Cayley's table
(ii) Find the order of the group
(iii) Find the identity element of the group
(iv) Find the order of each element in a group.
(v) Find the inverse of each element in a group.</td><td>10</td></tr>
<tr><td>ANS</td><td>Group
(i)
<table><tr><td>X_{40}</td><td>5</td><td>15</td><td>25</td><td>35</td></tr><tr><td>5</td><td>25</td><td>35</td><td>5</td><td>15</td></tr><tr><td>15</td><td>35</td><td>25</td><td>15</td><td>5</td></tr><tr><td>25</td><td>5</td><td>15</td><td>25</td><td>35</td></tr><tr><td>35</td><td>15</td><td>5</td><td>35</td><td>25</td></tr></table>(ii) order of the group is 4 (iii) identity element of the group is [25]
(iv) O[5]=2, O[15]=2,O[25]=2,O[35]=2
(v) inverse of [5] is [5], inverse of [15] is [15], inverse of [25] is [25], inverse of [35] is [35].</td><td></td></tr>
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NOTE*: Please refer below to the BL – Bloom's Taxonomy Levels and mention the respective level in the questions.

Bloom's Taxonomy Levels	**Category**
BL1	Remembering
BL2	Understanding
BL3	Applying
BL4	Analyzing
BL5	Evaluating
BL6	Creating